# Answers

## 1. Applications of Matrices and Determinants

### Exercise 1.1

1.(i) $\rho(A)=2$ (ii) $\rho(A)=2$ (iii) $\rho(A)=1$ (iv) $\rho(A)=3$

(v) $\rho(A)=2$ (vi) $\rho(A)=2$ (vii) $\rho(A)=3$ (viii) $\rho(A)=2$

2. $\rho(AB)=2, \rho(BA)=2$ 3. $x=1, y=3, z=5$

4. $x=\frac{1}{11}(7-16k)$, $y=\frac{1}{11}(3+k)$, $z=k$ 5. $x=2,\ y=1,\ z=0,\ ,$ 6. $\lambda=\frac{-7}{2}$

7. $x=1000,\ y=2000,\ z=500$ 8. $x=1000,\ y=2200,\ z=1800$

### Exercise 1.2

1.(i) $x=8\ ,\ y=-3$ (ii) $x=1,\ y=4$ (iii) $(x,\ y,\ z)=(2,\ -1,\ 0)$

(iv) $(x,\ y,\ z)=(1,\ 2,\ 3)$ (v) $(x,\ y,\ z)=\left(-1,\ \frac{1}{2},\frac{1}{3}\right)$

2. Cost per unit of labour is ₹10 Cost per unit of capital is ₹ 16

3. Amount invested at $4\frac{3}{4}\%$ is ₹7,300 Amount invested at $6\frac{1}{2}\%$ is ₹1,300

4. hourly charges for horse riding is ₹100 and AVT riding is ₹120

5. $(x,y,z)=(2,3,1)$

6. Amount invested at 2% is ₹250
   Amount invested at 3% is ₹4,000
   Amount invested at 6% is ₹4,250

### Exercise 1.3

1. 36% 2.(i) 54% , 46% (ii) 50%

3. A=56.25%, B=43.75% 4. A=33%,B= 67%



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